

Technical Requirements Document (TRD)

Ultra-High Accuracy Multilingual Real-Time Transcription Pipeline

1. Project Objective

To engineer a high-fidelity, zero-cost (Free Tier) universal transcription and translation engine. The primary goal is to achieve **near 100% accuracy** by eliminating mechanical chunking and utilizing a dual-stage inference pipeline that decouples acoustic recognition from linguistic translation.

2. The "Accuracy-First" Architecture

The system moves away from standard 3-second timers to **Semantic Integrity Enforcement**. The pipeline consists of three critical stages:

- **Acoustic Gating (Local Silero VAD):** Ensures audio is only processed upon completion of a natural sentence or phrase.
- **Primary Inference (Whisper Large-v3):** Performs high-resolution native transcription (capturing the speaker's original language).
- **Linguistic Refinement (Llama 3.3 70B):** Acts as a "Reasoning Layer" to correct ASR errors and provide context-aware translation.

3. Technical Specifications

A. Backend (Python/FastAPI)

- **VAD Framework:** silero-vad (Open Source).
- **Audio Handling:** 16kHz Mono PCM via WebSockets.

- **Inference Engine:** Groq LPU (Language Processing Unit) for ultra-low latency.

B. Dual-Inference Pipeline

Stage	Model	Role	Accuracy Impact
1. Acoustic	Whisper Large-v3	Native Transcription	Captures phonemes in original script without translation bias.
2. Linguistic	Llama 3.3 70B	Refinement & Translation	Fixes grammatical errors and maps technical jargon to the target language.

4. Engineering Requirements

Zero-Cost Strategy: All components utilize free-tier APIs (Groq) or local open-source models (Silero VAD).

- **Auto-Language Detection (ALD):** Automatic identification of the source language (English, Hindi, Spanish, etc.).
- **Contextual Preservation:** Technical terms (e.g., "API", "TensorFlow") are transliterated or preserved rather than literally translated.
- **Latency Target:** Total round-trip time $\Delta t < 1000ms$.

5. Antigravity Implementation Prompt

Use the following prompt within the Antigravity Agentic IDE to execute the project change:

```
"MISSION: Overhaul current project for NEAR 100% ACCURACY.  
1. Implement Silero VAD in a FastAPI backend to detect sentence  
boundaries via WebSockets.  
2. Chain Whisper Large-v3 (Transcription) to Llama 3.3 70B  
(Translation).  
3. Set Llama system prompt: 'Correct phonetic errors, identify  
source language, and translate to [TARGET] while maintaining  
technical context.'  
4. Update Chrome Extension to stream raw audio continuously."
```

6. Senior Defense (Project Justification)

*"Standard ASR tools fail because they destroy temporal context with fixed-time slicing. My architecture implements a **Semantic Buffer** using VAD, ensuring the transformer model receives a complete linguistic unit. By further decoupling transcription from translation, I utilize a 70B parameter LLM to provide error correction that standard Whisper models cannot achieve alone."*